

#1

```
name="python"
print(len(name))
print()
print(name[0])
print(name[1])
print(name[2])
print(name[3])
print(name[4])
print(name[5])
print()
print(name[-6])
print(name[-5])
print(name[-4])
print(name[-3])
print(name[-2])
print(name[-1])
```

#2

```
print()
print("list")
a=[1,2,3,4,5]
b=['a','e','i','o','u']
c=['neha',102,79.5]
d=["a b","e f","i j","o p","u v"]
e=['''a''','''e''','''i''','''o''','''u''']
f=['''ab''','''ef''','''ij''','''op''','''ur''']
g=['''a b''','''e f''','''i j''','''o p''','''u r''']
print(a,b,c,d,e,f,g)
```

#3

```
print()
print("tuple")
a=(1,2,3,4,5)
b=('a','e','i','o','u')
c=('neha',102,79.5)
d=("a b","e f","i j","o p","u v")
e=('''a''','''e''','''i''','''o''','''u''')
f=('''ab''','''ef''','''ij''','''op''','''ur''')
```

```
g=('a b','e f','i j','o p','u r')
print(a,b,c,d,e,f,g)
```

#4

```
print()
print("set")
a={1,2,3,4,5}
b={'a','e','i','o','u'}
c={'neha',102,79.5}
d={"a b","e f","i j","o p","u v"}
e={'a','e','i','o','u'}
f={'ab','ef','ij','op','ur'}
g={'a b','e f','i j','o p','u r'}
print(a,b,c,d,e,f,g)
```

#5

```
print()
print("dictionary")
a={1:1,2:2,3:3,4:4,5:5}
b={'a':1,'e':2,'i':3,'o':4,'u':5}
c={'neha':'a',102:'b',79.5:'c'}
d={"a b":1,"e f":2,"i j":3,"o p":4,"u v":5}
e={'a','e','i','o','u'}
f={'ab','ef','ij','op','ur'}
g={'a b','e f','i j','o p','u r'}
print(a,b,c,d,e,f,g)
print(c['neha'])
```

#6

```
print()
p=5
q=p
r=5
print(id(5))
print(id(p))
print(id(q))
print(id(r))
```

#7

```
print()
p=10
q=r
r=7
print(id(10))
print(id(p))
print(id(q))
print(id(r))
print(id(5))
```

#8

```
print()
chk=[2,4,6]
print(chk)
print(id(chk))
chk[1]=40
print(chk)
print(id(chk))
```

#9

```
print()
a=4
print(type(4))
print(type(a))
```

#10

```
print()
print("square")
a=4
b=2
print(a**b)
print("cube")
a=4
b=3
print(a**b)
print("sq root")
a=4
b=0.5
print(a**b)
```

```
print("cube root")
a=8
b=0.3333333333333333
print(a**b)
```

#11

```
print()
a,b,c,d=9.2,2.0,4,21
print(a/4,a//4)
print(b**c)
print(d//b)
print(a%c)
```

#12

```
print()
import math
r = 3.75
a = math.pi * r **2
print("area of Circle =", 'sq meters')
```

#13

```
print()
print(5//-3)
print(-5//3)
print(-5/3)
print(-5%3)
print(5%-3)
```

#14

```
print()
a=3
b=13
c='n'
d='g'
e='N'
f='god'
g='God'
h='god'
j='God'
```

```
k= 'Godhouse '  
L=[1,2,3]  
M=[2,4,6]  
N=[1,2,3]  
O=(1,2,3)  
P=(2,4,6)  
Q=(1,2,3)  
a<b  
c<d  
f<h  
f==h  
c==e  
g==j  
"God"<"Godhouse"  
"god"<"Godhouse"  
a==p  
L==M  
L==N  
O==P  
O==Q  
print(a==True)  
print(0==False)  
print(1==True)
```

#15

```
print()  
a=235  
b=240  
c=235  
print(a is b)  
print(a is c)  
b=b-a  
print(a is b)  
print(id(a),id(b),id(c))  
print(a,b)
```

#16

```
print()  
s1='abc'
```

```
s2= input("enter a string:")  
print(s1==s2)  
print(s1 is s2)  
s3='abc'  
print(s1 is s3)
```

#17

```
print()  
i=2+3.5j  
j=2+3.5j  
print(i is j)
```

#18

```
print()  
k=3.5  
l=float(input("Enter the value:"))  
print(k==1)  
print(k is 1)
```